CMPG 325 – COMPUTER NETWORKS

PROJECT MILESTONE 1: DESIGN SUBMISSION

Client: JB Marks Local Municipality Satellite Office (Ventersdorp)

Client ID: CLI-105

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1) Client Requirements Analysis

1.1) Client Overview

The client is the JB Marks Local Municipality Satellite Office located in Ventersdorp. This office operates within the Municipal Services industry and requires a computer network to support its daily administrative and public-facing operations.

1.2) Key Requirements

After analysing the project brief, the following key requirements were identified:

Connectivity: The network must provide appropriate connectivity and network services for the assigned scenario.

VoIP Separation: Voice over IP (VoIP) traffic must be strictly separated from standard data traffic to ensure clear call quality.

Internal DNS: The network must include an internal Domain Name System (DNS) server to resolve internal hostnames to IP addresses.

Public Kiosk (CR13): A public self-service kiosk must be installed at reception with internet-only access, blocking access to internal resources.

1.3) Addressing Constraint

The organization has been assigned the addressing block ***172.30.72.0/23***. All devices must fall within this range.

2) Physical Topology Design

2.1) Topology Selection:

The selected topology type is a **Hierarchical Star with Centralized Routing (also known as "Router-on-a-stick")** topology. This design uses a single main Router connected to a Core Switch, which then branches out to Access Switches for each department.

2.2) *Physical topology diagram*

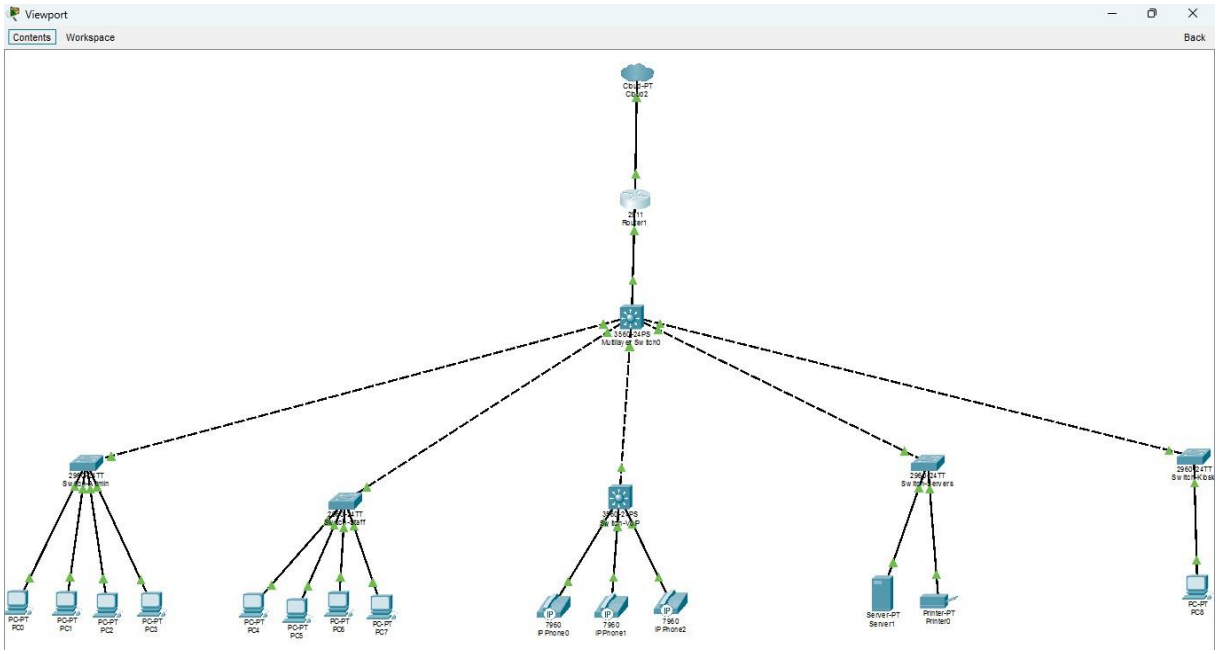


Figure 1: Physical topology showing all devices connected.

2.3) Device Inventory

1 x Router (2911): Acts as the main gateway and handles inter-VLAN routing.

1 x Core Switch (3560-24PS): Manages VLANs and distributes traffic.

5 x Access Switches (2960-24TT or 3560-24PS): Separate switches for Admin, Staff, VoIP, Servers, and Kiosk.

End Devices: 4 Admin PCs, 4 Staff PCs, 3 IP Phones, 1 DNS/DHCP Server, 1 Network Printer, and 1 Kiosk PC

3) Logical Topology Design

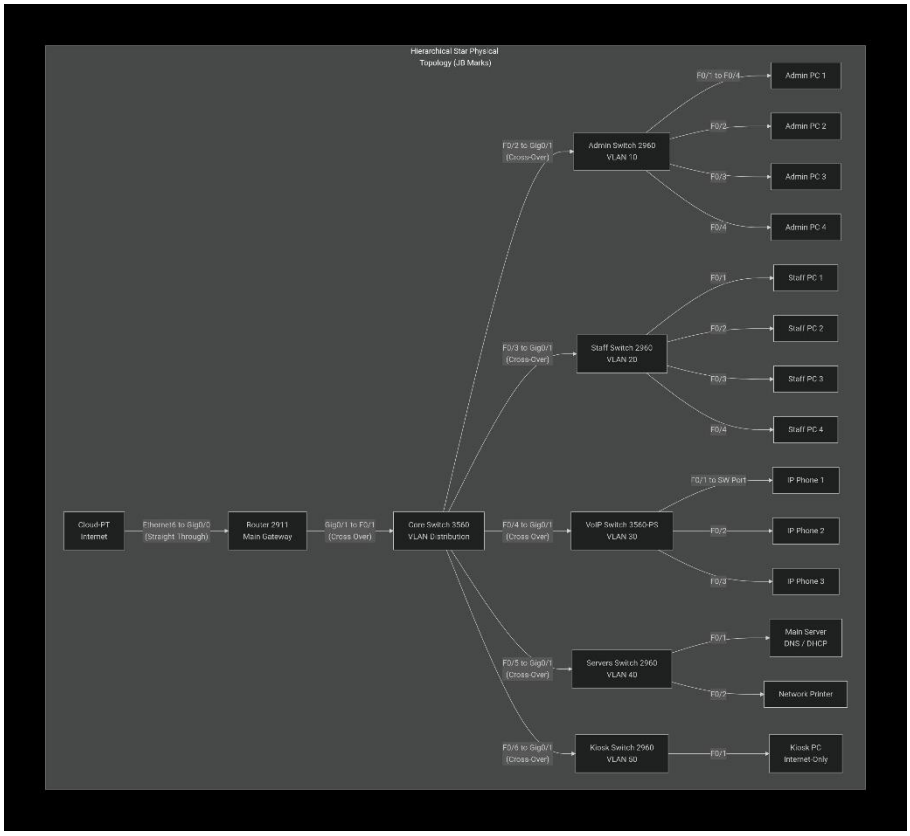


Figure 2: Logical topology

.1) *Subnet Overview:*

The addressing block ***172.30.72.0/23*** has been subnetted to create 5 separate logical networks. This ensures traffic isolation and improves security.

3.2) VLAN Mapping

Each physical subnet is mapped to a specific VLAN to separate the departments at the data-link layer.

4) IP Addressing Plan

Department / Subnet	VLAN ID	Network Address	CIDR	Subnet Mask	Usable IP Range	Gateway (Router)
Administration	VLAN 10	172.30.72.0	/25	255.255.255.128	172.30.72.1 – 172.30.72.126	172.30.72.1
Staff	VLAN 20	172.30.72.128	/25	255.255.255.128	172.30.72.129 – 172.30.72.254	172.30.72.129
VoIP (Phones)	VLAN 30	172.30.73.0	/26	255.255.255.192	172.30.73.1 – 172.30.73.62	172.30.73.1
Servers (DNS/DHCP/Printer)	VLAN 40	172.30.73.64	/28	255.255.255.240	172.30.73.65 – 172.30.73.78	172.30.73.65
Kiosk (Public)	VLAN 50	172.30.73.80	/28	255.255.255.240	172.30.73.81 – 172.30.73.94	172.30.73.81
(Future Expansion)	-	172.30.73.96	/27	255.255.255.224	172.30.73.97 – 172.30.73.126	

Table 1: Complete IP Addressing Plan

Key Device Assignments:

a) Main Server (DNS/DHCP): 172.30.73.66

b) Network Printer: 172.30.73.68

5) Design Justification

5.1) Why a Hierarchical Star?

The Hierarchical Star topology was chosen because it allows me to separate VoIP and Data traffic easily. It is also cost-effective and highly scalable. If the office grows, I can simply add another Access Switch to the Core Switch without redesigning the whole network.

5.2) Why separate subnets?

- **VoIP Isolation:** By placing IP Phones on **VLAN 30** ***(172.30.73.0/26)***, voice traffic does not compete with data downloads. This prevents call quality issues.

- **Kiosk Isolation**: Placing the public kiosk on **VLAN 50** ***(172.30.73.80/28)*** allows me to implement strict Access Control Lists (ACLs) on the Router. The Kiosk will be allowed to reach the Internet, but blocked from accessing the Admin, Staff, and Server networks.

- **Server Protection**: Placing the DNS/DHCP server on its own subnet adds a layer of security, making it harder for general staff or visitors to accidentally compromise the server.

5.3) Reason for subnet sizes

- **/25** for Admin/Staff provides 126 usable addresses. This allows for office growth.

- **/26** for VoIP provides 62 addresses, which is more than enough for a satellite office.

- **/28** for Servers and Kiosk provides 14 addresses each. These are specific services, so we don't want to waste IP addresses on large subnets for them.

6) Conclusion

After successfully analysing the client requirements and designing a physical and logical network that meets all requirements, including VoIP separation, DNS integration, and the public kiosk change request. The network is ready for implementation in Cisco Packet Tracer.

https://github.com/siyawnkosi29-debug/cmpg325-municipality-network